

## **Modeling of the Natural Gas HVOF Process**

**R. Bolot, M.P. Planche, C. Coddet,  
LERMPS-UTBM, Belfort, France**

### **Abstract**

A supersonic external flow and high particle velocities, which lead to low porosity coatings, characterize the High Velocity Oxygen Fuel (HVOF) process. A numerical study of this process is proposed in the present work. In a first step, the PHOENICS CFD code is used for the computation of the flow inside a CDS gun and within the supersonic external jet: a two-dimensional axisymmetric geometry is used. The combustion model assumes chemical equilibrium whereas the turbulence effect is taken into account using the Chen-Kim k- $\epsilon$  model, which is appropriate for the computation of round jets. In a second step, an in-house code is used in order to simulate the flow/particles interactions. Finally, numerical results obtained on in-flight particle characteristics are compared with measurements obtained with the DPV2000 system.

### **Introduction**

Industrial applications of the HVOF spraying technique increase for many years. Today, it is commonly used for wear resistance or applications against corrosion and oxidation [1]. Different industrial systems are available, including CDS, Jet Kote, Diamond Jet, D-Gun, Top-Gun or JP-5000. In all cases, burning gases are accelerated within a nozzle and the flow reaches a sonic or supersonic velocity at the nozzle exit. Most of the time, the supersonic external jet is under-expanded and contains luminous shock diamonds visible to the naked eye.

The main advantage of the process is the high velocity of the gases and then of the particles that leads to low porosity coatings with high adhesion: this is the reason for which these coatings may be used for wear and corrosion applications. Research developments focus on two different aspects: the first one concerns the system itself (combustion chamber, gun design) [1, 2], and the second one concerns the fuel nature [1, 3]. New developments require a better knowledge of the process including combustion phenomena, burning gas flow and gas/particles interactions. This can be achieved using

either diagnostic techniques or numerical simulations. Today, diagnostic systems such as the DPV2000 from TECNAR [4] constitute powerful tools for in-flight particle characteristic analysis.

Concerning modeling, many different studies have already been performed [1, 5-8]. Smith et al. [5] computed the internal flow and the external jet of the Diamond Jet process separately. In their jet study, they assumed that the flow at the nozzle exit was fully mixed and had reached chemical equilibrium. Oberkampf and Talpallikar [6] used the commercial code CFD-ACE and conducted a numerical study of the flow using an approximate equilibrium chemistry model (DJ gun too). Hassan et al. [7] extended the preceding works to realize a 3D modeling of the flow in the case of a rotating DJ gun. Yang [1] realized a complete modeling of the HVOF process for a CDS gun and established a numerical code composed of different modules [1, 8]. He performed two different calculations for the flow inside and outside the gun. However, a boundary layer approximation was used for the external jet model so that the possibility to take the presence of diamonds within the external jet into account was not provided. Moreover, the effect of turbulence was taken into account with a mixing length model.

The aim of this work is to present different improvements of Yang's works, especially concerning the external jet modeling that was the known weakness of the established code. A combustion model in which chemical equilibrium is assumed was implemented and the PHOENICS commercial CFD code was used for the modeling of the flow within the nozzle and the external jet simultaneously. Furthermore some numerical results concerning in-flight particle characteristics are presented and compared to measurements.

### **Natural Gas : Spray Parameters**

Yang et al. [1, 9] were the firsts to suggest using natural gas as fuel. Usually, other gases like propylene [6-8], propane, hydrogen, acetylene, ethylene, MAPP, ... are commonly

preferred. The authors sprayed different powders (WC-Co and Cr<sub>3</sub>C<sub>2</sub>-NiCr) with propane and natural gas successively. Many advantages were found in the use of natural gas (reliability, security, quality of the coatings produced), so that it is now widely used. The major constituent of natural gas is usually CH<sub>4</sub> (about 95%) whereas the other components depend strongly on the gas source. In the present study, the combustion of pure methane was considered as a first approximation.

A CDS gun from Sulzer-Metco with a 3 inches gun barrel was used for the spraying of a NiCrBFeSi powder (self-fluxing alloy). A CDS gun is composed of 3 different parts: <sup>(i)</sup> a converging chamber in which fuel, oxygen, the carrier gas and the powder are directly injected, <sup>(ii)</sup> a constant diameter barrel terminated by <sup>(iii)</sup> a slightly diverging zone (0.5°) in which the flow reaches supersonic conditions.

Table 1: Spraying conditions (stoichiometric conditions).

| CH <sub>4</sub> | O <sub>2</sub> | N <sub>2</sub> (carrier gas) |
|-----------------|----------------|------------------------------|
| 220 slpm        | 440 slpm       | 20 slpm                      |

### Basis of the model

#### Combustion model

A premixing of fuel, oxygen and nitrogen was assumed at the gun inlet. As it was said previously, the combustion model assumes chemical equilibrium. Thermodynamic properties of individual chemical species were taken from JANAF tables [10] and GURVICH's tables [11] which are very similar except that they do not use the same reference temperature. The considered species were those for which the concentration was higher than 0.1% over the whole or a part of the temperature and pressure range. Thus, the retained species were: N<sub>2</sub>, O<sub>2</sub>, H<sub>2</sub>, H<sub>2</sub>O, CO<sub>2</sub>, CO, NO, O, OH and H. The method used for the computation of chemical equilibrium is a Gibbs Free Energy minimization method based on the RAND algorithm: details concerning this method may be found in [1, 12-13]. Figure 1 presents the computed chemical equilibrium composition of the combustion products versus temperature at a pressure of 5 bars (absolute).

At low temperature, the combustion products contain CO<sub>2</sub>, H<sub>2</sub>O and N<sub>2</sub> mainly, but the concentrations of the dissociated species increase with the temperature increase. Since stoichiometric conditions were used, the oxygen content is low at low temperature.

The concentrations of the species being known versus the temperature, one can deduce (for a given pressure) the adiabatic combustion temperature defined as the temperature at which the enthalpy of the initial reagents equals the enthalpy of the combustion products. In the present case, a very good fitting of the adiabatic combustion temperature was obtained by:

$$T_C = 3043 P_C^{0.04}$$

where  $T_C$  (K) is the combustion temperature and  $P_C$  (bar) is the absolute combustion pressure.

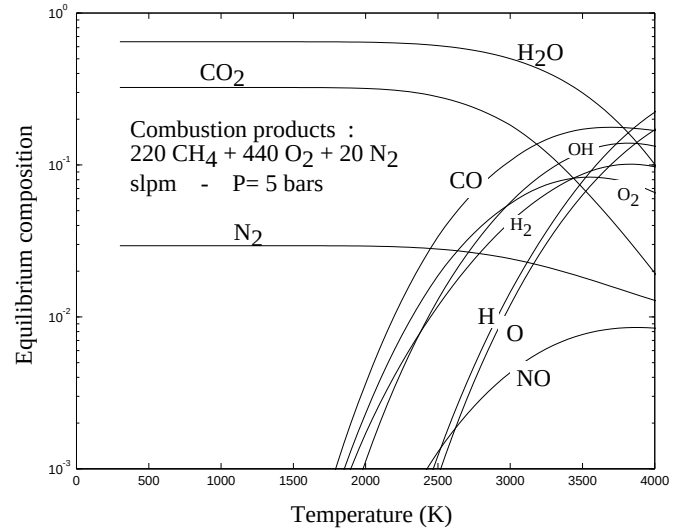


Figure 1: Combustion product composition versus temperature at a pressure of 5 bars (absolute).

#### Thermodynamic properties and transport coefficients

Thermodynamic properties can be directly deduced from the chemical equilibrium composition. Transport coefficients were calculated following the kinetic theory of gases. The Wilke mixing rule [14] was used for the computation of the dynamic viscosity of the combustion gas mixture (and that of the surrounding air) whereas its counterpart of Mason & Saxena [15] was used for the thermal conductivity. By doing so, the calculated thermal conductivity does not include the contribution of chemical reactions (it is the frozen thermal conductivity which is thus calculated). Consequently, the equilibrium thermal conductivity was estimated from  $\lambda_e/\lambda_f \approx C_{p_e}/C_{p_f}$  [16] as first approximation where indices <sub>e</sub> and <sub>f</sub> stand for equilibrium and frozen respectively. Considering the results of Yang [1], it seems that the use of this method for the calculation of  $\lambda_e$  produces an underestimation of the contribution of chemical reactions. Consequently, one can suppose that  $\lambda_e$  was certainly weakly underestimated in the present study (especially at high temperature for which the contribution of chemical reactions becomes important).

Concerning interaction potentials (required to calculate transport coefficients: e.g. dynamic viscosity and thermal conductivity), the Lennard-Jones potential was used for each interaction. This potential is certainly the most popular and the most used for the calculation of transport properties of combustion products. Consequently, parameters are accessible in the literature for a large number of species [17-18].

Figure 2 presents the equilibrium and frozen specific heat for the same mixture as described above and for two different pressure levels. One can observe that the equilibrium specific heat becomes significantly higher than the frozen one for temperature beyond 2000 K, corresponding to an important amount of energy stored chemically. The pressure influence is also significant: the lower the pressure, the higher the importance of the reactive part for temperatures lower than the combustion one.

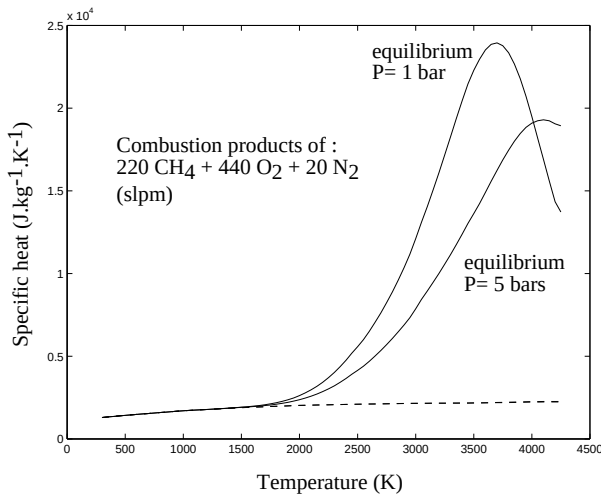


Figure 2: Specific heat of the combustion products versus temperature for two different pressure levels.

#### CFD model

Yang [1] used a MacCormack predictor/corrector scheme to compute the flow inside the gun. By doing so, external conditions are extrapolated at the nozzle exit. He also built two different programs to compute the external flow but could not take the shock waves simultaneously with dispersion into account.

In the present work, the use of a single program to compute the internal flow and the external one simultaneously is suggested: the PHOENICS<sup>(1)</sup> code is particularly adapted and provides this possibility. A finite volume method based on the SIMPLEST discretization algorithm is embodied within the software. The possibility to use different "collocated velocities" options allows users to consider highly non-orthogonal grids. Several efficient turbulence models (k- $\epsilon$ , RSTM, ...) are directly available. Moreover, most of the high order schemes (QUICK, SMART, TVD, bounded schemes, ...) are also implemented and can be applied to both scalars and velocities. In the present study, we used the GCV method (Generalized Collocated Velocities) to perform the computations since it was found to be more efficient than the default staggered grid method. Concerning turbulence modeling, we used the Chen-Kim k- $\epsilon$  model [19-20] since it is by far better adapted (versus the standard one) for the computation of round jets. Finally, the high order scheme by Van Albada and Van Leer [21] was used for the discretization of convective terms, allowing a good capture of shock waves within the external jet and preventing numerical diffusion.

#### Mixing with the surrounding air

In the case of plasma jets, the nature of the environment is known to have a decisive influence on the cooling of the jet. In view of this phenomenon, we used a model similar to that

described in [13, 20] to take the mixing with air into account. In this model, a conservation equation is added for the mass fraction of the primary gas (combustion products) and mixing rules are used to estimate the properties of the mixture (combustion products + surrounding air). The energy equation includes two different diffusive contributions: (i) heat conduction due to temperature gradients and (ii) the contribution of inter-diffusion of the two mixtures. Thermodynamic properties and transport coefficients of the combustion products and air are therefore necessarily calculated previously and separately. For this, tabulated data were used and a three parameters interpolation procedure was coded within PHOENICS (temperature, pressure and combustion products mass fraction).

Figures 3 and 4 present the dynamic viscosity and the frozen thermal conductivity of the combustion products and air at the surrounding pressure. One has to notice that the equilibrium thermal conductivity is much higher in view of the use of  $\lambda_e/\lambda_f \approx C_{p_e}/C_{p_f}$  (see Fig. 2).

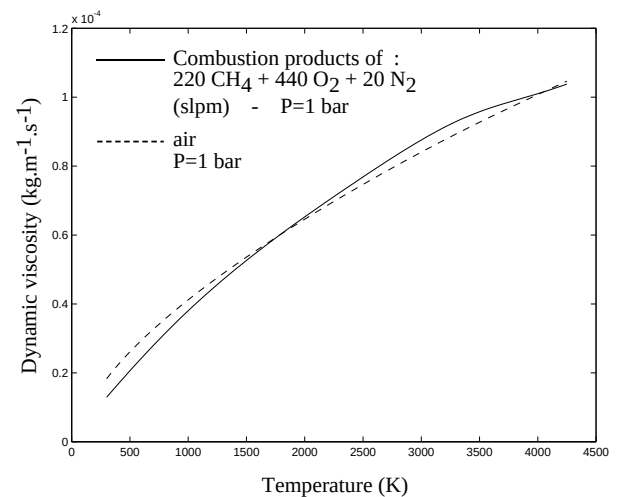


Figure 3: Dynamic viscosity of the combustion products and of the surrounding air.

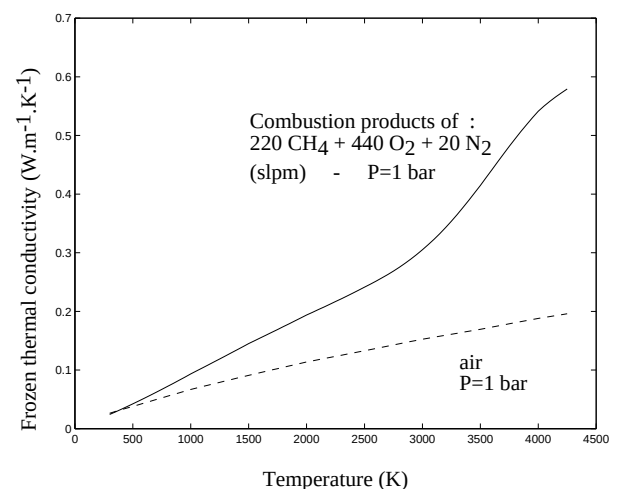


Figure 4: Frozen thermal conductivity of the combustion products and of the surrounding air.

<sup>1</sup> PHOENICS is a general purpose CFD software commercialized by CHAM, Bakery House, 40 High Street, Wimbledon Village, LONDON SW 19 5AU, UK.

## Numerical Results

### Axial profiles

Figure 5 presents the computed axial profiles of the pressure. The upwind pressure was determined using an iterative method in order to fit the effective mass flow rate. The pressure drops quickly within the converging chamber, more slowly along the constant area duct, and quickly again within the diverging part of the nozzle. Nevertheless, the pressure remains higher than 1.5 bar (absolute) at the nozzle exit so that the presence of waves is predicted along the external jet.

Figure 6 presents the Mach number axial profile. The Mach number increases quickly within the convergent and more slightly along the constant area duct. The sonic transition appears at the inlet of the divergent. At this point ( $M=1$ ) the corresponding velocity is about 1200 m/s. The Mach number reaches a value of about 1.15 at the nozzle exit and increases up to 1.8 within the jet at the point where the pressure is the smallest. The mean Mach number in the core of the external jet is about 1.5 whereas the corresponding velocity is about 1800 m/s.

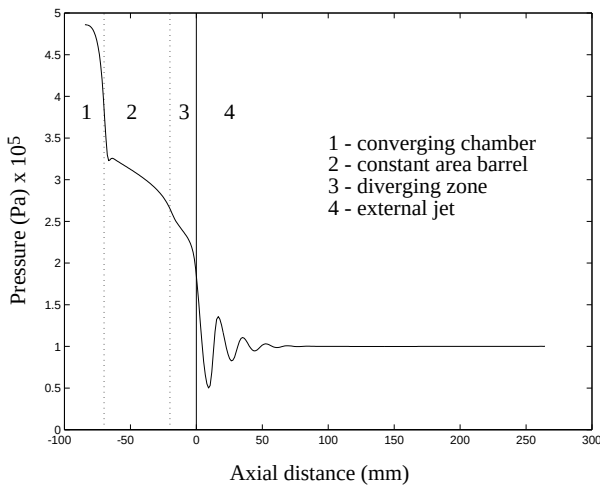


Figure 5: Axial pressure along the centerline of the flow.

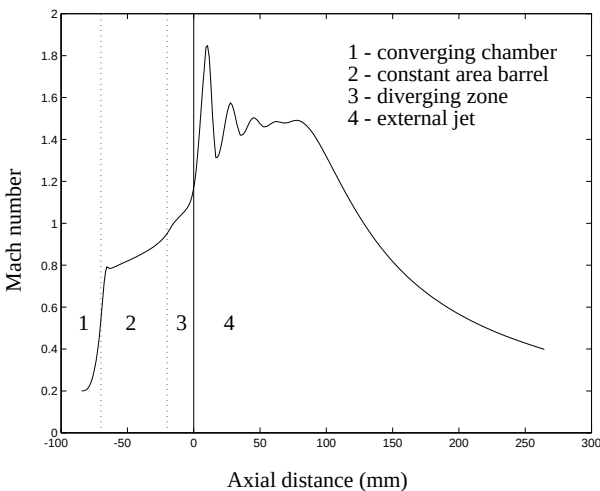


Figure 6: Mach number along the centerline of the flow.

Figure 7 presents the temperature profile along the centerline. The temperature drops along the nozzle but remains higher than 3000 K at the gun exit. This has to be correlated to the equilibrium assumption: the temperature drop associated with the pressure drop is small since chemical recombination reactions occur and maintain a high temperature level. This phenomenon is also traduced through the  $\gamma$  coefficient: the increase in the equilibrium specific heat at high temperature produces a decrease of  $\gamma$  which is about 1.15 around 3000 K. Assuming an isentropic flow along the nozzle, the temperature ratio  $T/T_0$  would be directly correlated to the corresponding pressure ratio through

$$\frac{T}{T_0} = \left( \frac{P}{P_0} \right)^{\frac{\gamma-1}{\gamma}}$$

A decrease of  $\gamma$  provides a decrease of the temperature ratio. This is also the reason for which the magnitude of the temperature waves observed within the external jet are also small. Meanwhile, this region presents high gradients so that the chemical equilibrium assumption could be discussed (the  $\gamma$  coefficient is higher for a frozen situation). High-pressure regions and high temperature regions along the waves are superimposed and so more luminous, so that diamonds may be visible to the naked eye.

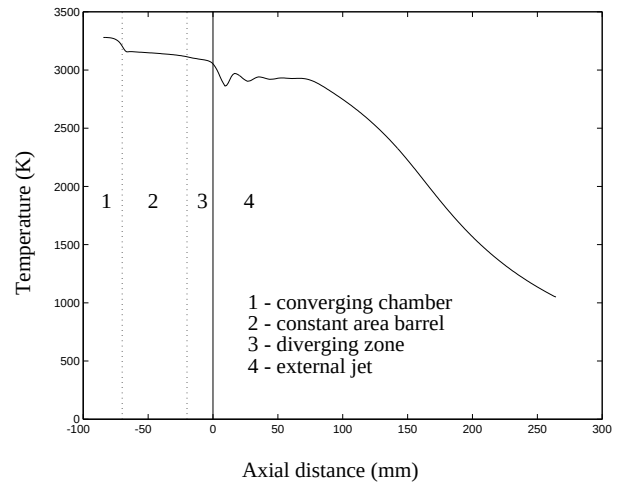


Figure 7: Temperature along the centerline of the flow.

### In-Flight Particles

A particle tracking model was also implemented and allows the prediction of particle velocities and temperatures within the flow. The DPV 2000 sensing device [4], based on the detection of thermal radiation of hot particles, was used to perform in-flight measurements of particle velocities and temperatures at different locations within the external jet. Consequently, the comparison of modeling and measurement results provides information about the validity of the model. Figure 8 presents the particle size distribution of the initial powder in terms of volume and number fraction taken from the sieving analysis: the mean particle size is about 39  $\mu\text{m}$  in terms of volume fraction and about 32  $\mu\text{m}$  in terms of number

fraction. Particles smaller than 20  $\mu\text{m}$  were not taken into account in the mean particle size in terms of number fraction: in fact, even if the volume fraction of the smallest particles is weak, their contribution can be significant in terms of number. However, the smallest particles are not detected by the DPV 2000.

Some of the thermodynamic properties of the NiCrBFeSi alloy were estimated from the properties of the major constituents of the powder (density, specific heat, latent heat) whereas the other (melting point and thermal conductivity) were taken in the literature. The presence of Boron and Silicon provides a lowering of the melting point (950°C-1100°C) whereas the thermal conductivity of NiCr alloys is much smaller than that of pure Ni or pure Cr [22]. Table 2 presents the composition of the NiCrBFeSi alloy.

Figure 10 presents the calculated velocity of particles of different size along the centerline of the flow. The smaller is the particle, the higher its velocity. The predicted velocity for the largest particles (65  $\mu\text{m}$ ) is higher than 300 m/s, whereas it is as high as 600 m/s for the smallest particles (20  $\mu\text{m}$ ). For a standoff distance of 280 mm, the mean particle velocity measured with the DPV 2000 system was 450 m/s with a standard deviation of about 50 m/s. Moreover, one can expect that this measured mean velocity is that of the mean particle size in terms of number fraction (that is to say about 32  $\mu\text{m}$ ). According to the numerical results, the particle velocity is higher than the jet one at the standoff distance. Consequently, all the particles reached their maximum velocity and are not accelerated anymore at this distance. In view of the particle size distribution in terms of number fraction, one could think that the calculated particle velocities are somewhat overestimated. However, all the particles do not follow the centerline (where the jet velocity is the higher) but travel at different radial distances from the axis.

Thus, figure 11 presents the acceleration of particles that would not follow the centerline but would be seen at a radial position of 10 mm at the standoff distance. These numerical results are more in accordance with DPV measurements in view of the particle size distribution.

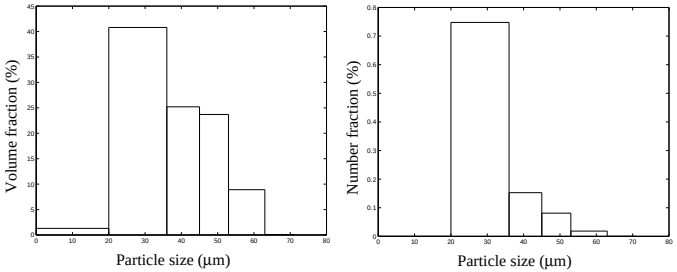


Figure 8: Particle size distribution in terms of volume and number fraction (NiCrBFeSi powder).

Table 2: Composition of the NiCrBFeSi powder.

| Ni  | Cr  | B    | Fe   | Si   | C     |
|-----|-----|------|------|------|-------|
| 73% | 15% | 3.2% | 3.8% | 4.3% | 0.75% |

In view of fig. 10 and 11, the computed particle velocity is sensitive to the radial position in the jet. However, one has to notice that this was not the case during experiments: the measured particle velocity remains almost constant when the volume control is displaced in the radial direction.

Concerning temperatures, some difficulties remain due to the particle oxidation that occurs during the spraying. This phenomenon may have a significant influence on the particle temperature. Consequently, the comparison between numerical and experimental results may be biased for the temperature. Moreover, numerical results indicate a fast particle heating in the converging chamber where the gas velocity remains low and where the temperature is the higher (since the premixing of fuel and oxygen was assumed). Consequently, an overestimation of thermal exchanges may be obtained in the converging part of the nozzle where in fact the oxygen-fuel mixing and combustion occurs.

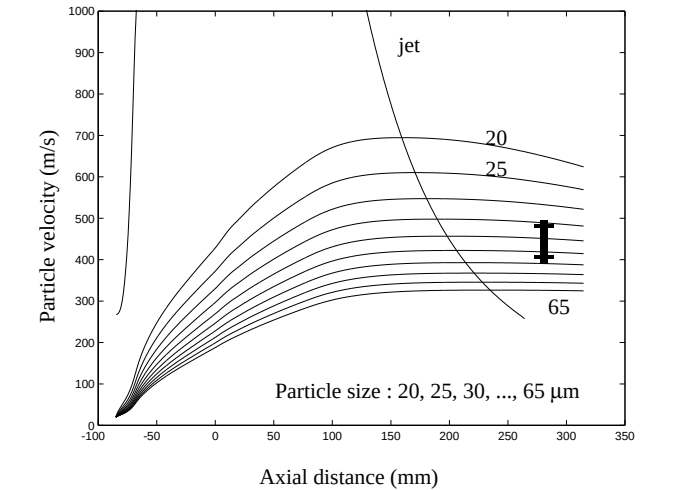


Figure 10: Calculated velocities along the centerline for particles of different size. The stroke represents the measured velocities (mean particle size = 32  $\mu\text{m}$ ).

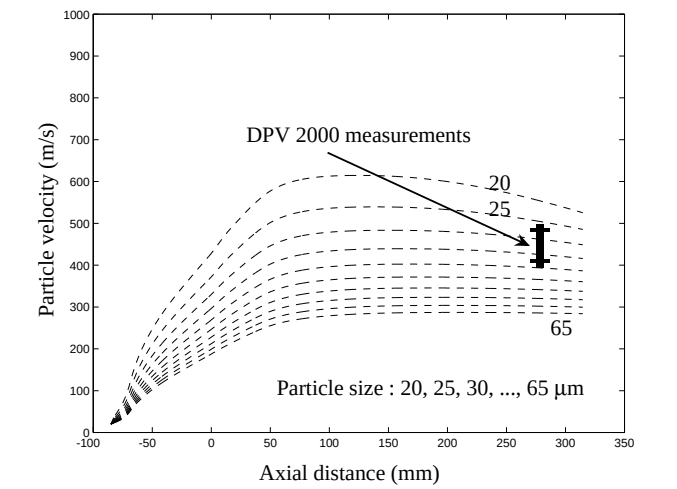


Figure 11: Calculated velocities for particles of different size at 10 mm from the centerline. The stroke represents the measured velocities (mean particle size = 32  $\mu\text{m}$ ).

## Conclusions

The present paper was devoted to a modeling of the HVOF spraying process. Thermodynamic properties of the combustion products of natural gas were determined on the basis of the chemical equilibrium assumption. The PHOENICS commercial CFD code was used for the computation of the flow inside and outside a CDS gun simultaneously. The sonic transition appears at the exit of the barrel but, since the diverging angle is small, the flow remains under-expanded and shock waves are observed in the external jet. The use of an extended computational domain and a round jet adapted turbulence model provides a good idea of the turbulent dispersion of the jet. The process of mixing with the surrounding air was also taken into account. Finally, the predicted and measured particle in-flight velocities compare satisfactorily and reinforce the validity of the current model.

## References

1. Y.M. Yang, "Mathematical Modeling of a HVOF Thermal Spray System: Application to the Development of Protecting Coatings", PhD Thesis, University of Franche-Comté, France, 1996 (in French).
2. K. Sakaki, Y. Shimizu, Y. Gouda, and A. Devasenapathi, "Effect of gun nozzle geometry on high velocity oxygen-fuel (HVOF) thermal spraying process", *Thermal Spray: Meeting the Challenges of the 21<sup>st</sup> Century*, Ed. C. Coddet, Pub. ASM Int., Materials Park, OH, 1998, pp. 445-450.
3. V. Gourlaouen, E. Verna, and J. Sattonnet, "Comparison of different fuel gases in HVOF", *Thermal Spray: Meeting the Challenges of the 21<sup>st</sup> Century*, Ed. C. Coddet, Pub. ASM Int., Materials Park, OH, 1998, pp. 861-866.
4. C. Moreau, P. Gougeon, M. Lamontagne, V. Lacasse, G. Vaudreuil, and P. Cielo, "On-line Control of the Plasma Spraying Process by Monitoring the Temperature, Velocity and Trajectory of in-flight Particles", *Thermal Spray Industrial Applications*, Ed. C.C. Berndt and S. Sampath, Pub. ASM Int., Materials Park, OH, 1994, pp. 431-437.
5. E.B. Smith, G.D. Power, T.J. Barber, and L.M. Chiappetta, "Application of Computational Fluid Dynamics to the HVOF Thermal Spray Gun", *Thermal Spray: International Advances in Coatings Technology*, Ed. C.C. Berndt, Pub. ASM Int., Materials Park, OH, 1992, pp. 805-810.
6. W.L. Oberkampf, and M. Talpallikar, "Analysis of a High Velocity Oxygen-Fuel (HVOF) Thermal Spray Torch - Part 1: Numerical Formulation", *Thermal Spray Industrial Applications*, Ed. C.C. Berndt and S. Sampath, Pub. ASM Int., Materials Park, OH, 1994, pp. 381-386.
7. B. Hassan, A.R. Lopez, and W.L. Oberkampf, "Computational Analysis of a Three-Dimensional High-Velocity Oxygen-Fuel (HVOF) Thermal Spray Torch", *Advances in Thermal Spray Science & Technology*, Ed. C.C. Berndt and S. Sampath, Pub. ASM Int., Materials Park, OH, 1995, pp. 193-198.
8. Y.M. Yang, C. Coddet, and M. Imbert, "Numerical modeling and simulation of particle behaviors in an HVOF supersonic flow", in *Thermal Spray: Meeting the Challenges of the 21<sup>st</sup> century*, Ed. C. Coddet, Pub. ASM Int., Materials Park, OH, 1998, pp. 431-437.
9. Y.M. Yang, H. Liao, and C. Coddet, "New Performance for HVOF Thermal Spraying Systems with the Use of Natural Gas", in *Thermal Spray: Current Status and Future Trends*, Ed. A. Ohmori, Pub. High Temp. Society of Japan, 1995, Vol. 1, pp. 307-312.
10. M.W. Chase et al., "JANAF Thermochemical Tables", *J. Phys. and Chem. Reference Data*, 14, Suppl. 1 (1985).
11. L.V. Gurvich, I.V. Veyts, and C.B. Alcock, "Thermodynamic Properties of Individual Substances", 4<sup>th</sup> edition, Vol. 1, Hemisphere (1989).
12. W.R. Smith, and R.W. Missen, "Chemical Reaction Equilibrium Analysis: Theory and algorithms", John Wiley & Sons (1982).
13. R. Bolot, "Modeling of Burning Arc Plasma Flows: Application to the Spray of Powder Materials", PhD Thesis, University of Franche-Comté, France, 1999 (in French).
14. C.R. Wilke, "A viscosity equation for gas mixtures", *The Journal of Chemical Physics*, 18(4) (1950) 517-519.
15. E.A. Masson, and S.C. Saxena, "Approximate Formula for the Thermal Conductivity of Gas Mixtures", *The Physics of Fluids*, 1(5) (1958) 361-369.
16. J.N. Butler, and R.S. Brokaw, "Thermal Conductivity of Gas Mixtures in Chemical Equilibrium", *The Journal of Chemical Physics*, 26(6) (1957) 1636-1643.
17. J.O. Hirschfelder, C.F. Curtiss, and R.B. Bird, "Molecular theory of gases and liquids", 4<sup>th</sup> edition, John Wiley & Sons (1967).
18. R.A. Svehla, "Estimated Viscosities and Thermal Conductivities of Gases at High Temperatures", NASA Technical Report TR-132, Lewis Research Center, OH, 1961.
19. Y.S. Chen, and S.W. Kim, "Computation of Turbulent Flows using an Extended k- $\epsilon$  Turbulence Closure Model", NASA CR-179204, 1987.
20. R. Bolot, M. Imbert, and C. Coddet, "On the Use of a Low-Reynolds Extension to the Chen-Kim k- $\epsilon$  Model to Predict Thermal Exchanges in the case of an Impinging Plasma Jet", *International Journal of Heat & Mass Transfer*, 44(6) (2001) 1095-1106.
21. G.D. Van Albada, B. Van Leer, and W.W. Roberts, "A Comparative Study of the Computational Methods in Cosmic Gas Dynamics", *J. Astron. Astrophysics*, 108 (1982) 76.
22. *Handbook of Chemistry and Physics*, 75<sup>th</sup> Edition, Ed. D.R. Lide, CRC Press (1995), Chapter 12-163.